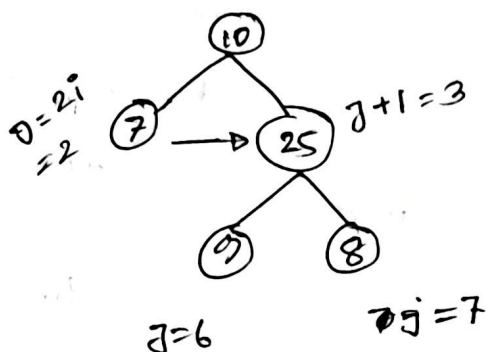
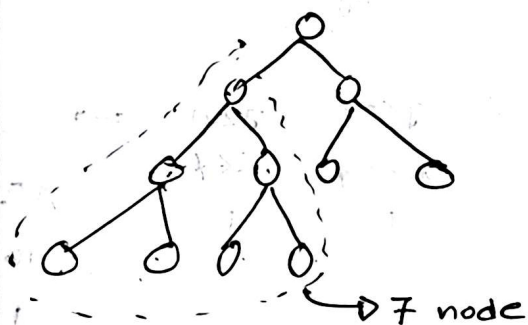




** in complete binary tree there should be maximum $\frac{2n}{3}$ node in any sub tree.

if $A[j] \leq item$
 ~~$A[6] \leq$~~
 $(9 \leq 10)$ True
 exit loop.

Time complexity Analysis: (max heapify)



$$n=11$$

$$\frac{2n}{3} = \left\lfloor \frac{2 \times 11}{3} \right\rfloor \text{ (Subtree)}$$

$$= 7$$

Recurrence relation for time, $T(n)$

$$T(n) = \begin{cases} 0 & n=1 \\ T\left(\frac{2n}{3}\right) + c & n > 1 \end{cases}$$

Adjust time

for comparison left/right

(*)